

Logistics Package Tracking System

Project Report

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Abstract

This project presents a Java-based Logistics Package Tracking System developed using Object-Oriented Programming (OOP). The system allows customers to register, book parcels, generate unique tracking IDs, update shipment status, and track parcels throughout the delivery process.

Objectives

- Develop a logistics package tracking application.
- Manage customer registration and login.
- Generate unique tracking IDs.
- Track parcel status and location.
- Demonstrate Object-Oriented Programming concepts.

Introduction

Logistics and courier companies require efficient systems to monitor parcel movement and delivery. This project simulates a logistics tracking platform where customers can book parcels, logistics representatives update shipment details, and customers track deliveries using a unique tracking ID.

System Modules

- Register – Customer registration and authentication.
- CustomerBooking – Parcel booking and tracking ID generation.
- Parcel – Stores parcel details.
- RepresentativeUpdate – Updates parcel location and delivery status.
- CustomerTracking – Displays parcel information using tracking ID.

- MainSystem – Controls application workflow and menu navigation.

System Workflow

1. Customer registers an account.
2. Customer logs into the system.
3. Customer books a parcel.
4. System generates a tracking ID.
5. Representative updates parcel location and status.
6. Customer tracks the parcel using the tracking ID.

OOP Concepts Used

- Classes and Objects
- Encapsulation
- Modular Programming
- Method Invocation
- Object Communication

Algorithms

Registration and Login

Parcel Booking

Tracking ID Generation

Parcel Search

Status Update

Customer Tracking

Technologies Used

Java

Object-Oriented Programming

Scanner Class

Random Number Generation

Console-based Programming

Applications

- Courier Services
- Logistics Companies
- Supply Chain Management
- Delivery Tracking
- Warehouse Management

Future Scope

- Database integration using MySQL
- JavaFX/Swing graphical interface
- SMS and Email notifications
- GPS-based live tracking
- Multiple parcel management
- Admin dashboard

Conclusion

The Logistics Package Tracking System demonstrates how Object-Oriented Programming can be used to build a modular logistics management application. It provides customer registration, parcel booking, shipment tracking, and representative updates through a simple console-based interface.

References

1. Oracle Java Documentation
2. Object-Oriented Programming Concepts
3. Java Scanner Class Documentation